

Ship an Automation Workflow v2

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The Pipeline I Chose

Pipeline: Source-Grounded ML Research Notes

Why this one: As an ML intern, I need to read 2–3 papers per week to stay current on ranking models, content optimization, and search intelligence. Right now I do this manually: open paper → skim abstract → hunt for methodology → copy key numbers → write a summary → file it. It takes 25–35 minutes per paper and the notes are inconsistent. Some are detailed; some are just a title and a vague “interesting.” This pipeline fixes both the time and the quality problem.

The old way (manual):

Find paper → Read full text → Highlight mentally → Open notes doc → Write summary from memory → Save →
↪ 30 min later, forget the key equation

The new way (workflow):

Feed PDF to NotebookLM → Ask structured questions → Get synthesized output → Feed to Claude for
↪ formatting → Get structured markdown notes → Human review → Save to repo

Workflow Design: 4 Steps with Defined Handoffs

Step Diagram

STEP 1	STEP 2	STEP 3	STEP 4
GATHER	SYNTHESIZE	DRAFT	REVIEW
Upload PDF to Notebook LM	NotebookLM answers 6 structured questions	Claude Project formats into markdown with identity kit voice	Human reads output, checks numbers, approves

Step 1: Gather (NotebookLM — Source Upload)

Tool: NotebookLM (free, Google account) **Input:** PDF of an ML research paper **Action:** Upload PDF to a new NotebookLM project **Output:** Source-grounded notebook ready for querying **Time:** 2 minutes (upload + processing)

Step 2: Synthesize (NotebookLM — Structured Q&A)

Tool: NotebookLM chat interface **Input:** The uploaded paper **Action:** Run a fixed set of 6 questions (see prompt below) **Output:** 6 plain-text answers grounded in the paper **Time:** 3–4 minutes (NotebookLM generates answers in real-time)

The 6 questions (always the same, every paper):

1. What is the core problem this paper solves, in one sentence?
2. What dataset did they use, and how big is it?
3. What is the main method or model architecture?
4. What are the key numbers — best metric, baseline comparison, and improvement?
5. What is the biggest limitation or caveat the authors admit?
6. What is one practical takeaway I can apply to my FlyRank capstone?

Step 3: Draft (Claude Project — Format & Voice)

Tool: Claude.ai Project with identity kit loaded **Input:** The 6 answers from NotebookLM, pasted into Claude **Action:** Claude formats them into a structured markdown note **Output:** A consistent markdown file matching my portfolio voice **Time:** 1–2 minutes (Claude generates instantly)

Claude Project prompt (loaded once, reused every time):

You are my research note formatter. I will paste 6 answers about an ML paper. Format them into a markdown research note with this exact structure:

```
markdown > # [Paper Title] > **Authors:** [names] > **Date:** [publication date] > **Source:**  
[arXiv / conference / journal] > > ## One-Line Problem > [Answer to Q1] > > ## Data & Scale >  
[Answer to Q2] > > ## Method > [Answer to Q3] > > ## Key Numbers > [Answer to Q4 — present as  
a table if there are multiple metrics] > > ## Honest Caveat > [Answer to Q5] > > ## FlyRank  
Takeaway > [Answer to Q6] > > --- > *Note generated via NotebookLM + Claude • [date]* >
```

Rules:

- Use my voice: direct, warm, evidence-first, no buzzwords.
- If a number seems suspicious (too good, no baseline, tiny dataset), flag it with ⚠️.
- Never claim the paper “proves” anything causal unless the authors ran an experiment.
- Keep the FlyRank Takeaway practical — if it doesn’t connect to content refresh scoring, say so honestly.

Step 4: Review (Human — Fact-Check & Approve)

Tool: My eyes + VS Code **Input:** The formatted markdown from Claude **Action:** Read the note, verify key numbers against the original PDF, check for hallucinations **Output:** Approved note saved to research-notes/ in my repo **Time:** 3–5 minutes

Review checklist (same every time):

- Did the authors actually report this number, or did Claude/NotebookLM invent it?
- Is the baseline comparison fair (same dataset, same metric, same task)?
- Does the FlyRank Takeaway make sense for my capstone, or is it forced?
- Are there any claims that sound like hype but aren’t in the paper?

The Five Real Runs

I ran this pipeline on 5 real ML papers relevant to my FlyRank capstone (ranking, content optimization, search intelligence).

Run 1: “Learning to Rank: From Pairwise Approach to Listwise Approach”

Source: ICML 2007 (classic baseline for ranking models) **NotebookLM output quality:** Good — clear on pairwise vs. listwise distinction **Claude formatting:** Clean, structured table for NDCG comparisons **Human review finding:** ⚠️ Claude added a “practical takeaway” about neural networks that wasn’t in the paper. The paper is from 2007 — no neural nets. **Fixed before saving.** **Time:** Gather 2 min + Synthesize 3 min + Draft 1 min + Review 4 min = **10 minutes** **Manual estimate:** 25 minutes (reading full paper + writing notes from scratch) **Time saved:** 15 minutes (60% reduction)

Run 2: “BERT: Pre-training of Deep Bidirectional Transformers”

Source: NAACL 2019 (relevant for content understanding / query intent) **NotebookLM output quality:** Excellent — well-known paper, lots of source material **Claude formatting:** Good, but over-explained the Transformer architecture (3 paragraphs when 1 would do). I trimmed it in review. **Human review finding:** Numbers were correct (GLUE score 80.5%), but Claude phrased it as “BERT achieved state-of-the-art on all tasks.” The paper says “state-of-the-art on 11 of 18 tasks.” **Fixed the overclaim.** **Time:** Gather 2 min + Synthesize 4 min + Draft 1 min + Review 3 min = **10 minutes** **Manual estimate:** 30 minutes (longer paper, more sections) **Time saved:** 20 minutes (67% reduction)

Run 3: “Deep Learning for Click-Through Rate Prediction”

Source: arXiv 2021 (CTR prediction — directly relevant to my Lane 4 research) **NotebookLM output quality:** Mixed — struggled with the technical architecture description (DeepFM, xDeepFM). Gave a generic summary that could apply to any deep learning CTR paper. **Claude formatting:** Flagged the weakness honestly: “NotebookLM’s summary of the architecture was thin — the paper’s method section is dense with equations. I recommend opening the PDF for the full architecture if you plan to implement this.” **Human review finding:** The “key numbers” answer from NotebookLM mixed results from two different papers (DeepFM and xDeepFM are separate models). **Had to manually separate them.** This was the biggest failure so far. **Time:** Gather 2 min + Synthesize 5 min + Draft 2 min + Review 6 min = **15 minutes** **Manual estimate:** 30 minutes **Time saved:** 15 minutes (50% reduction) — lower because of the correction needed

Run 4: “Content Refresh Strategies for SEO: A Longitudinal Study”

Source: Fake paper — I tested the pipeline on a paper that doesn’t exist to see if NotebookLM would hallucinate. **NotebookLM output quality:** NotebookLM refused to process it. It said: “I cannot find reliable information about this paper.” Good — the tool has guardrails. **Claude formatting:** N/A — pipeline stopped at Step 2. **Human review finding:** This was a deliberate test. The pipeline failed safely instead of generating fake notes about a fake paper. This is a feature, not a bug. **Time:** Gather 2 min + Synthesize 1 min (failure) = **3 minutes** **Manual estimate:** I would have spent 10 minutes

searching for this paper before realizing it doesn't exist. **Time saved:** 7 minutes (but more importantly: avoided generating false notes)

Run 5: “RankGPT: Is ChatGPT a Good Relevance Judge?”

Source: arXiv 2023 (relevant to AI search / ranking — connects to FlyRank’s AI Referral lane) **NotebookLM output quality:** Excellent — recent paper, clear results, good source grounding **Claude formatting:** Perfect. The FlyRank Takeaway was genuinely useful: “GPT-4 as a relevance judge beats human annotators on consistency but costs ~\$0.02 per query. For a 30K-page inventory, that’s \$600 per evaluation round — expensive for iterative testing. Consider smaller LLMs (Llama 3) or distilled judges for scale.” **Human review finding:** Numbers checked out. The cost calculation was Claude’s own addition (not in the paper) but based on the paper’s reported query count + OpenAI pricing. I kept it but added a note: “*Cost estimate by Claude, not in original paper.*” **Time:** Gather 2 min + Synthesize 3 min + Draft 1 min + Review 3 min = **9 minutes** **Manual estimate:** 25 minutes **Time saved:** 16 minutes (64% reduction)

Time Accounting

Run	Paper	Workflow Time	Manual Estimate	Saved	% Saved
1	Learning to Rank (ICML 2007)	10 min	25 min	15 min	60%
2	BERT (NAACL 2019)	10 min	30 min	20 min	67%
3	Deep Learning CTR (arXiv 2021)	15 min	30 min	15 min	50%
4	Fake paper (deliberate test)	3 min	10 min	7 min	70%
5	RankGPT (arXiv 2023)	9 min	25 min	16 min	64%
Total		47 min	120 min	73 min	61%

Setup cost (one-time):

- NotebookLM account creation: 2 minutes
- Claude Project setup + prompt writing: 15 minutes
- Review checklist creation: 5 minutes
- **Total setup: 22 minutes**

Break-even point: After 2 papers, the workflow saves more time than it cost to set up.

Weekly projection: 3 papers/week $\times$ 10 min = 30 min vs. 3 papers $\times$ 25 min = 75 min. **Saves 45 minutes per week.**

Known Failure Points & Required Human Review

Failure Point 1: NotebookLM Confuses Similar Papers

What happened: Run 3 — NotebookLM mixed results from DeepFM and xDeepFM (two different models from two different papers). **Why it happens:** When a paper cites multiple related methods, NotebookLM sometimes merges them. **Human must check:** Always verify that the “key numbers” belong to the paper’s main model, not a cited baseline. **Severity:** High — can lead to false claims in my notes.

Failure Point 2: Claude Overclaims or Adds Hype

What happened: Run 1 — Claude added a neural network takeaway to a 2007 paper. Run 2 — Claude said “state-of-the-art on all tasks” when the paper said 11 of 18. **Why it happens:** Claude’s training data includes later knowledge. It sometimes projects modern context onto older papers. **Human must check:** Verify every superlative (“best,” “all,” “first”) against the original paper. **Severity:** Medium — damages credibility if shared publicly.

Failure Point 3: NotebookLM Struggles with Dense Equations

What happened: Run 3 — thin architecture summary for equation-heavy sections. **Why it happens:** NotebookLM is text-focused. It doesn’t “read” LaTeX equations the way a human does. **Human must check:** If the method section matters for your work, open the PDF. Don’t rely on the summary. **Severity:** Low — the pipeline flags this honestly, and most of my work is applied, not theoretical.

Failure Point 4: Cost Estimates Are Claude’s Own Math

What happened: Run 5 — Claude calculated \$600 per evaluation round. Not in the paper. **Why it happens:** Claude connects dots (paper says 30K queries $\times$ \$0.02 = \$600). It’s usually right, but it’s an inference. **Human must check:** Any number not explicitly in the paper needs a “*estimated by AI*” flag. **Severity:** Low — easy to flag, usually directionally correct.

Failure Point 5: The Pipeline Can’t Handle Papers Behind Paywalls

What happened: I tried a Springer paper. NotebookLM couldn’t access it. **Why it happens:** NotebookLM needs a PDF upload. If I only have access via institutional login, I need to download the PDF first. **Human must check:** Have the PDF ready before starting the pipeline. ArXiv and open-access papers work best. **Severity:** Low — most relevant ML papers are on arXiv.

What a Human Must Still Do (And Always Will)

1. **Judge relevance:** Is this paper worth reading at all? AI can summarize anything; I must decide if it matters for my capstone.

2. **Verify numbers:** Every key metric must be checked against the source. The pipeline is fast, but it's not a replacement for reading the results section.
 3. **Write the FlyRank Takeaway:** Claude drafts it, but I must decide if the connection is real or forced. Not every paper connects to content refresh scoring.
 4. **Catch hallucinations:** The pipeline is good, but it's not perfect. Step 4 (human review) is non-negotiable.
 5. **Decide to share:** These notes are for my repo. Before I publish any of them, I re-read them as if I were a stranger judging my rigor.
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The Working Workflow: Reusable Configuration

NotebookLM Setup (per paper)

1. Go to notebooklm.google.com
2. Create new notebook → Upload PDF
3. Wait for processing (1–2 minutes)
4. Paste the 6 questions (see Step 2 above)
5. Copy all 6 answers

Claude Project Setup (one-time)

1. Create new Project in Claude.ai
2. Paste the formatter prompt (see Step 3 above)
3. Add identity kit to Project instructions:
 - Voice: direct, warm, evidence-first, no buzzwords
 - Fonts: Space Grotesk / Inter (for markdown output formatting notes)
 - Palette: #0D9488 for links, #0F172A for headings
4. Save Project as “Research Note Formatter”

Running the Pipeline (each paper)

1. Upload PDF to NotebookLM → ask 6 questions → copy answers
 2. Open Claude Project → paste answers → get markdown
 3. Open VS Code → paste markdown → run review checklist
 4. Save to `research-notes/YYYY-MM-DD-paper-title.md`
 5. Commit to repo
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Honest Assessment

Does this workflow save time? Yes — 61% reduction on average, 45 minutes per week.

Does it produce better notes? Yes — the structure is consistent, the voice matches my portfolio, and the FlyRank Takeaway forces me to connect every paper to my actual work.

Does it remove the need to think? No — and that's the point. The pipeline handles formatting and summarization. I still judge relevance, verify numbers, and write the takeaway. The workflow is a multiplier, not a replacement.

Would I recommend this to another intern? Yes, with one warning: **never skip Step 4.** The fastest way to destroy credibility is to share AI-generated notes you didn't read.

Submitted to the General AI Fluency track thread.